



Operando X-ray transmission microscopy of dynamic concentration change of electrolyte salt in composite electrodes of lithium-ion battery

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ABSTRACT

Spatiotemporal evolution of electrolyte salt concentration in lithium-ion battery composite electrodes during charge-discharge is investigated using synchrotron-based *operando* X-ray transmission microscopy. A custom-designed *operando* electrochemical cell enables cross-sectional imaging of the anode and separator regions under fast charge-discharge battery operating conditions. By exploiting the absorption contrast of the PF_6^- anion, relative changes in electrolyte salt concentration are visualized in real time. During high-rate discharge, lithium ions are released from the anode, leading to transient salt accumulation inside the porous electrode and the formation of pronounced concentration gradients. In contrast, concentration variations across the separator are smaller and relax more rapidly. The measurements further reveal that such concentration inhomogeneities are strongly rate-dependent and persist on time scales comparable to fast cycling, but gradually relax once the current is stopped. These results demonstrate that electrolyte concentration polarization dynamically develops during charge-discharge and can significantly influence ionic transport and cell performance, highlighting the critical role of electrolyte transport properties in high-rate lithium-ion batteries.

1. Introduction

Lithium-ion batteries (LIBs) employing high-loading composite electrodes are a promising strategy to increase energy density, particularly for electric vehicles and large-scale energy storage applications. However, these thick electrodes face significant limitations during fast charge-discharge operations, primarily due to transport-related polarization phenomena [1–3]. Typically, composite electrodes are fabricated by casting a slurry of active material, conductive carbon, and polymer binder onto a current collector, followed by drying and pressing. Within such electrodes, local reaction rates vary depending on ionic resistance (electrolyte conductivity, charge-transfer resistance at interfaces, and solid-state diffusion) and electronic resistance (through the active particles, conductive additives, and interparticle contacts) [4–7]. Both internal factors—such as electrolyte viscosity, conductivity, and pore structure—and external factors like temperature and pressure affect ion and electron transport pathways within the electrode, resulting in spatially heterogeneous reaction distributions during operation [8–15].

During battery operation, lithium ions undergo dynamic redistribution, resulting in spatial inhomogeneities in both the solid and liquid phases of the cell. Specifically, two distinct types of lithium distribution become relevant during battery operation: (1) the local lithium content in the active material particles, which reflects the state of charge of the solid phase, and (2) the local electrolyte salt concentration in the electrolyte-filled pores of the composite electrode.

In the initial state, both distributions are typically uniform, but significant spatial gradients can develop during high-rate or prolonged cycling. For the solid phase, spatial variations in lithium content across the electrode thickness arise from transport limitations and non-uniform reaction kinetics, leading to heterogeneous states of charge [16–18]. Such inhomogeneities have been investigated using optical microscopy [19,20], X-ray absorption spectroscopy [21,22], X-ray diffraction [16, 23–26], and other spatially resolved techniques [27], revealing that lithium insertion often proceeds preferentially from the separator side. These gradients contribute to localized overpotentials, reduced rate capability, and accelerated degradation.

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In contrast, the second type of inhomogeneity—the electrolyte salt concentration in the electrolyte—has received much less direct experimental attention, despite its critical role in governing ionic transport, especially under high-rate conditions. The electrolyte fills the pores of both the electrodes and the separator and serves as the main pathway for lithium-ion migration. During discharge, lithium ions are released from the anode and inserted into the cathode, thereby redistributing the electrolyte salt concentration and causing local salt depletion in the electrolyte within the cathode porous electrode due to mass-transport limitations.

Owing to the limited ionic diffusivity and viscosity of standard liquid electrolytes, significant concentration gradients can develop not only across the separator but also inside the electrodes—particularly in thick electrodes operated at high current densities. Numerical simulations based on porous-electrode models predict that, during charging, the electrolyte salt concentration becomes lowest near the anode/separator interface, where lithium-ion consumption is most intense [28–30]. Although such gradients are challenging to observe directly, *operando* techniques such as nuclear magnetic resonance (NMR) imaging [31,32] and X-ray phase contrast imaging [33] have provided direct evidence of electrolyte depletion, albeit in simplified systems without full composite electrodes or separators. Despite the recognized importance of electrolyte concentration polarization, direct *operando* visualization of spatio-temporal salt concentration changes within both the electrode and separator regions remains largely unexplored.

In this study, we address this gap by developing an *operando* X-ray transmission imaging method to track relative electrolyte salt concentration across the anode and separator layers of a lithium-ion battery cell. By leveraging the X-ray absorption contrast of the PF_6^- anion in a LiPF_6 -ethylene carbonate (EC) and diethyl carbonate (DEC) electrolyte, we visualize real-time concentration changes during charge–discharge cycling with spatial resolution. This technique enables the direct observation of how concentration gradients form, evolve, and relax under practical battery operating conditions.

2. Experimental methods

Active materials of cathode and anode were commercially available $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (NCM) and synthetic graphite, respectively. The cathode slurry was prepared by mixing 94 wt % NCM, 3 wt % acetylene black, and 3 wt % polyvinylidene difluoride (PVDF) binder in N-methyl-2-pyrrolidone (NMP), and coated onto aluminum foil. The anode slurry contained 98 wt % graphite and 2 wt % PVDF, coated onto copper foil. Both electrodes were vacuum-dried and roll-pressed. A 20 μm thick polypropylene separator was placed between the electrodes. 1.0 M LiPF_6 in a 3:7 vol ratio of EC and DEC was used as the electrolyte solution.

A custom-designed *operando* electrochemical cell was developed to enable cross-sectional X-ray transmission imaging. The composite electrodes used in this study had a coating thickness of approximately 50 μm , which is comparable to that of practical electrodes. The electrode width along the X-ray beam direction was set to approximately 1 mm to ensure sufficient transmitted X-ray intensity. The schematic of the cell structure is shown in Fig. 1a, with photos provided in Supplementary Fig. S1. All cell assembly processes were carried out in an argon-filled glovebox. The assembled cell was sealed with aluminum laminate films to allow safe handling under ambient conditions.

Galvanostatic charge–discharge tests were conducted at room temperature with voltage limits set between 3.0 and 4.1 V. Despite the reduced electrode area, the *operando* cell demonstrated a capacity comparable to that of practical cells, as shown in Fig. 1b, confirming the relevance of the observed phenomena to real battery operation.

Operando X-ray transmission microscopy measurements were performed at beamline BL33XU of SPRING-8 (Hyogo, Japan) using X-ray energy of 15 keV [34]. The beam size was 1 mm $\times$ 1 mm. X-ray transmission images were captured using CMOS detector (ORCA-Flash4.0, Hamamatsu Photonics) with LuAG scintillator. The integration time was

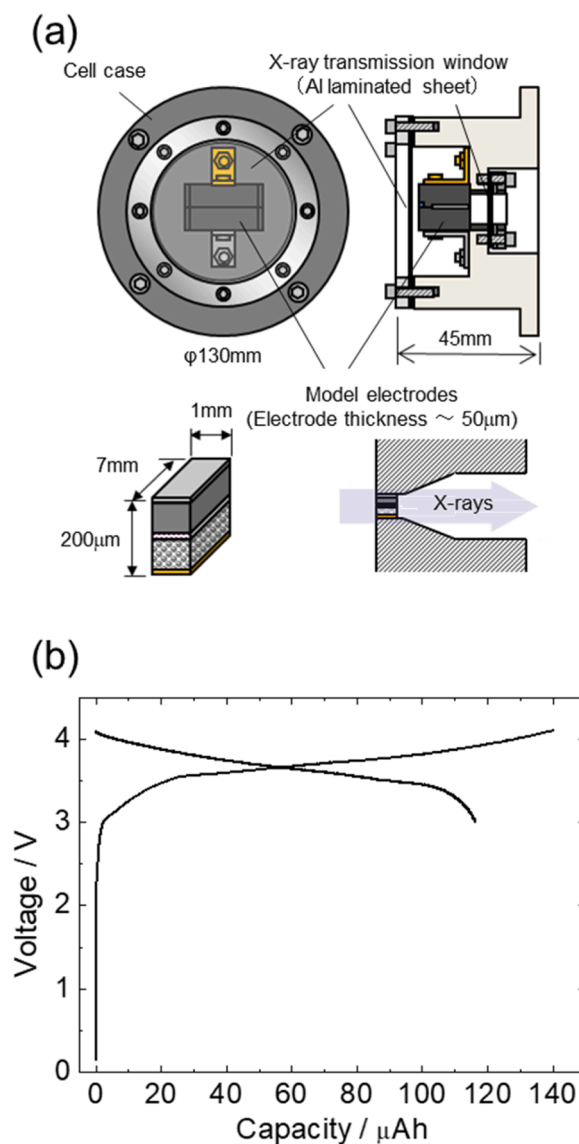


Fig. 1. (a) Schematic illustration of *operando* battery cell in this study. (b) Charge and discharge voltage profiles of *operando* cell at current density corresponding to 1C rate.

0.1 s per frame, and the pixel size was 1 $\mu\text{m} \times 1 \mu\text{m}$. X-ray transmission images were acquired during discharge–charge at 10C, 5C and 3C rate. The X-ray energy was selected to optimize contrast based on PF_6^- absorption, allowing detection of salt concentration changes across approximately 1 mm of electrolyte path length.

3. Results and discussions

To investigate how electrolyte salt concentration changes during battery operation, we employed *operando* X-ray transmission microscopy to visualize spatial variations within the anode and separator regions. Fig. 2a shows the X-ray transmission image of the cross-sectional view of the assembled *operando* cell. Distinct features corresponding to the aluminum current collector, cathode layer, separator, anode, and copper current collector can be observed due to the differences in X-ray transmittances among materials. However, because the cathode contains 3d transition metals that strongly absorb X-rays at 15 keV, our analysis focuses on the separator and anode regions where the X-ray beam penetrates effectively.

To visualize electrolyte concentration changes during operation, we

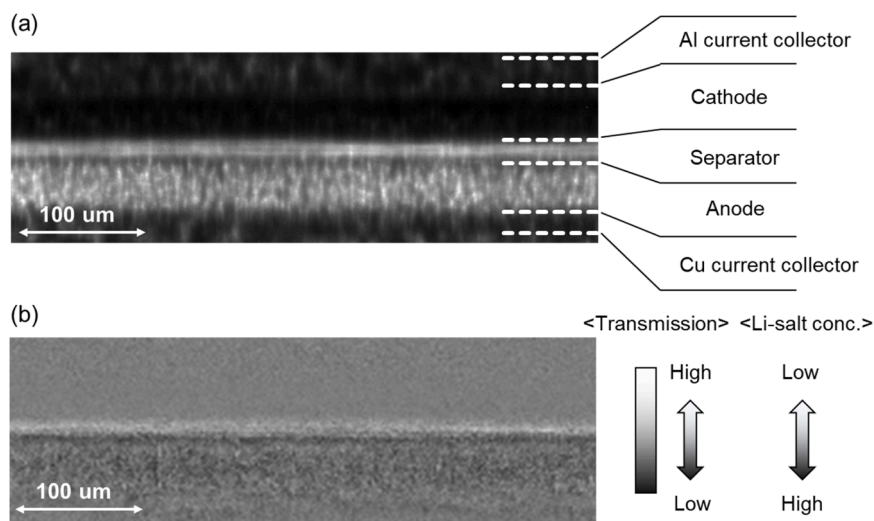


Fig. 2. (a) X-ray transmission image of the cross-section of the *operando* cell, showing layers corresponding to the anode, separator, and cathode. (b) Differential image obtained by subtracting the initial image from that captured after 30 s of discharge at 10C. The intensity decrease (darker region) corresponds to increased PF_6^- concentration, used as an indicator of salt distribution changes in the electrolyte.

obtained differential images by subtracting the initial transmission image from those recorded during battery operation. It should be noted that the present analysis focuses on relative changes in X-ray transmission intensity, and does not provide an absolute quantification of salt concentration. The measured contrast is therefore interpreted in terms of qualitative and concentration variations. An example of such a difference image captured after 30 s of 10C discharge is provided in Fig. 2b. Because Li has negligible X-ray absorption at this energy, the observed contrast is dominated by PF_6^- and thus reflects changes in electrolyte salt concentration. Assuming local electroneutrality, variations in PF_6^- concentration can therefore be interpreted as an indicator of lithium salt concentration changes in the electrolyte [31,32]. Because X-ray absorption is dominated by heavier elements, the contribution of lithium to the transmission change is negligible at 15 keV, whereas variations in PF_6^- concentration lead to significantly larger changes, indicating that the observed contrast primarily reflects changes in electrolyte salt concentration. In addition, by analyzing differential images obtained at identical positions under *operando* conditions, the influence of static structural features and potential uncertainties such as beam inhomogeneity is effectively minimized. To extract spatial profiles, the X-ray transmission signal was integrated along the horizontal direction, yielding one-dimensional intensity distributions across the separator and anode layers. These profiles were used to evaluate electrolyte concentration distributions during discharge and charge.

Charge–discharge reactions lead to changes in electrolyte salt concentration not only within the separator layer but also inside the porous composite electrodes. Fig. 3 shows the change in X-ray transmission intensity across the cross-section of the cell, alongside corresponding voltage profiles during 10C discharge and charge. During discharge (Fig. 3a and b), the intensity in both the anode and separator regions gradually decreases compared to the initial state. Assuming local charge neutrality, a decrease in X-ray intensity reflects an increase in PF_6^- concentration, indicating higher local lithium salt concentration in the electrolyte. Within the anode, a progressive intensity reduction is observed as discharge proceeds, reflecting an increase in electrolyte salt concentration. The intensity drop is more pronounced at 30 s than at 15 s, suggesting that lithium ion extracted from the anode is not fully diffused across the separator under high-rate conditions, and that the resulting concentration distribution is not relaxed on this timescale.

An unrelaxed concentration inhomogeneity develops within the composite electrode during high-rate operation. As shown in Fig. 3b, the X-ray transmission intensity is higher near the separator side than in the

middle region across the electrode thickness, indicating a non-uniform salt concentration. This inhomogeneity may arise from two factors. First, the reaction distribution across the electrode thickness tends to be non-uniform, with electrochemical reactions proceeding preferentially near the separator side, as previously reported by several groups [16,21, 25]. Second, the relatively slow diffusion of electrolyte salt in the electrolyte within the porous electrode structure delays the homogenization of salt concentration. Consequently, the lithium salt concentration remains elevated in the middle of the electrode during high-rate discharging. These results indicate that both reaction localization and slow ionic diffusion in the electrolyte contribute to the observed concentration inhomogeneity within the composite electrode.

A concentration gradient is also formed in the separator layer. At the early stage of discharge, the lithium salt concentration becomes higher near the anode side than near the cathode side. This gradient arises because lithium ions are released from the anode and consumed at the cathode during discharge, and mass transfer in the electrolyte is not sufficiently fast to homogenize the concentration. Similar behavior has been observed using *operando* NMR imaging [31,32], although the NMR cells in those studies did not contain a separator, but rather a pure electrolyte layer. In addition, the observed concentration change is governed by ionic migration and diffusion, which are strongly influenced by the lithium-ion transference number and diffusion coefficient [32]. In the limiting case of solid electrolytes, where the lithium-ion transference number approaches unity, such concentration gradients are not expected to develop [35]. Furthermore, electrolyte composition, such as salt concentration and salt type, may affect the behavior through changes in transport properties and viscosity. A systematic investigation of these effects is left for future work.

Electrolyte concentration gradients caused by mass transport gradually relax under open-circuit conditions once the driving force is removed. Fig. 3c shows the change in X-ray transmission intensity during open-circuit rest following a 10C discharge. The intensity profile progressively flattens, approaching its pre-discharge state after approximately 20 s of rest. These results suggest that such electrolyte concentration changes primarily occur during high current density charge–discharge reactions and contribute to a temporary decrease in local ionic conductivity.

During the charging process, the concentration changes in both the composite electrode and the separator layer exhibit the opposite trend to that observed during discharge, as shown in Fig. 3d–f. In this case, lithium ions are consumed at the anode side, leading to a decrease in the

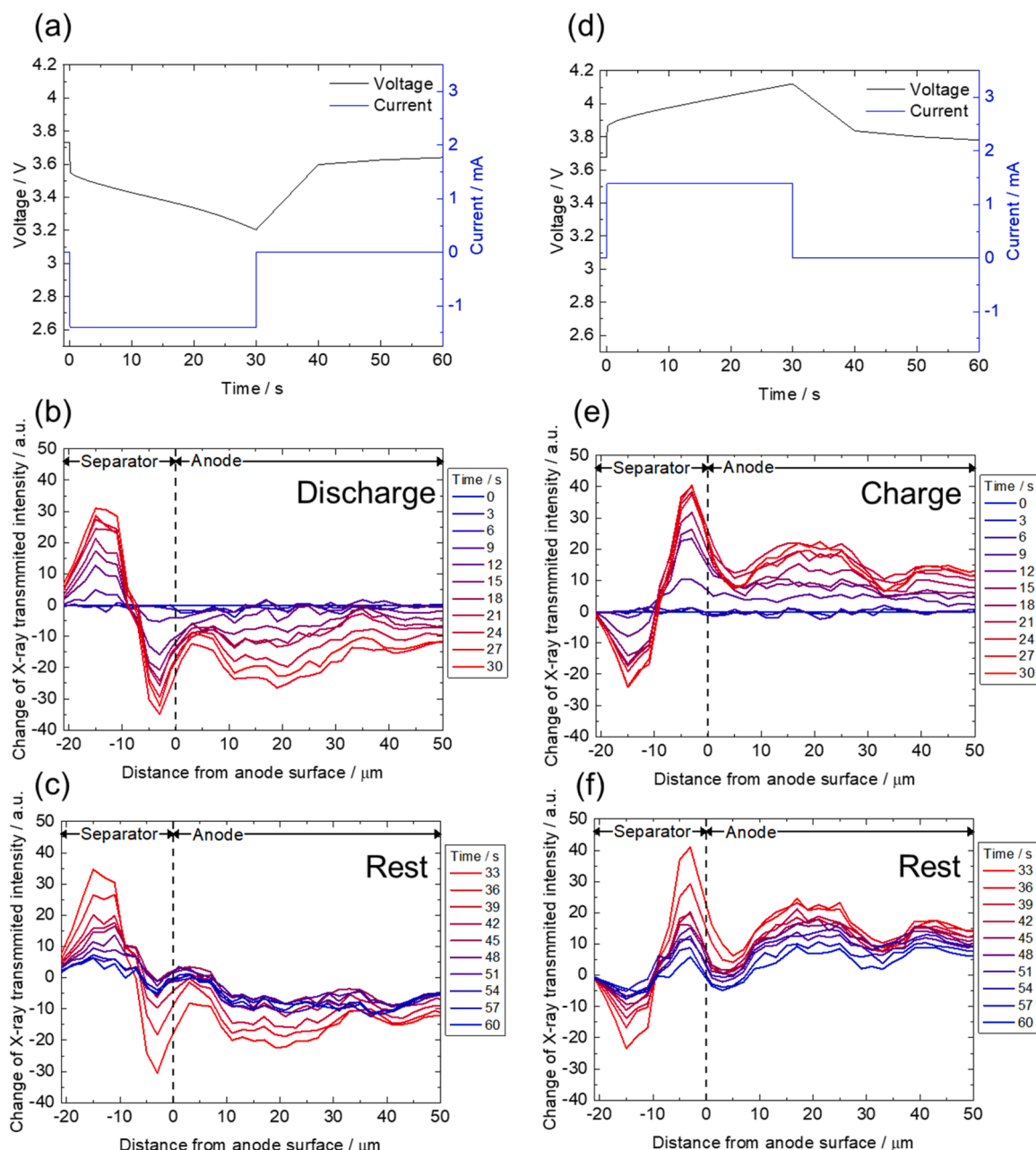


Fig. 3. Voltage profiles and corresponding X-ray transmission intensity changes across the separator and anode regions at a 10C rate. (a) Discharge voltage and current profiles. (b) Change in X-ray transmission intensity during discharge, relative to the initial state. (c) Change in X-ray transmission intensity during open-circuit rest after discharge, relative to the initial state. (d) Charge voltage and current profiles. (e) Change in X-ray transmission intensity during charge, relative to the initial state. (f) Change in X-ray transmission intensity during open-circuit rest after charge, relative to the initial state. In panels (b), (c), (e), and (f), lower X-ray intensity corresponds to higher PF_6^- concentration and thus higher local lithium salt concentration in the electrolyte.

electrolyte salt concentration within the anode. A concentration gradient also forms in the separator layer, with a lower salt concentration observed near the anode side.

The observed concentration gradient is strongly influenced by the current density, as it originates from the limited diffusivity of lithium ions in the electrolyte. Fig. 4a and b compare the X-ray transmission intensity changes across the cell cross-section during discharge and charge at 10C, 5C, and 3C, under similar capacity conditions. (The series of data are shown in Figs. S2, S3 and S4.) As previously observed, discharge and charge produce opposite trends in intensity variation. The decrease in intensity in the anode—reflecting lithium salt accumulation—is most prominent at 10C and becomes progressively less pronounced at 5C and 3C. This clearly indicates that the applied current is

the primary driver of concentration gradients, while electrolyte concentration changes are minimal at lower rates. Notably, although the concentration gradient across the separator becomes negligible at 3C, distinct intensity variations persist within the anode. This implies that electrolyte concentration relaxation is slower within the composite electrode than in the separator, highlighting the critical importance of electrode design in mitigating transport limitations.

Based on our observations, Fig. 5 schematically illustrates the evolution of lithium salt concentration within the composite electrode during battery operation. During the discharge reaction, lithium ions are extracted from the anode and released into the electrolyte, leading to a transient accumulation of lithium salt inside the porous electrode. The relaxation of such concentration gradients is governed not by the self-

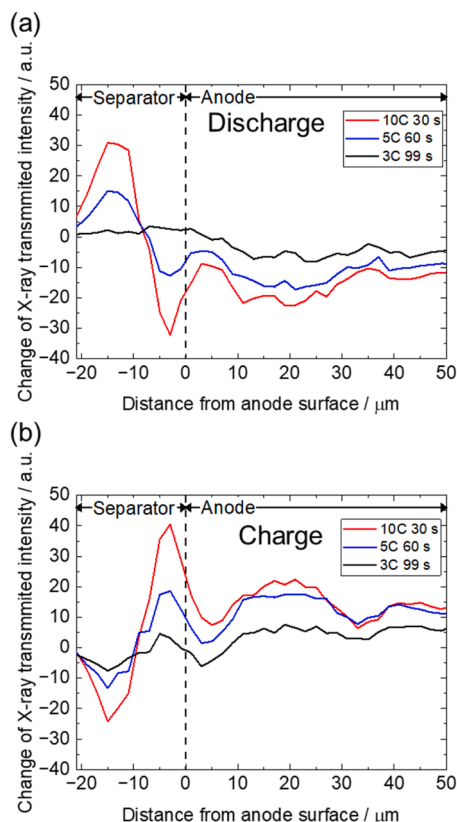


Fig. 4. X-ray transmission intensity changes across the separator and anode regions during (a) discharge and (b) charge at 10C, 5C, and 3C, for similar delivered capacities.

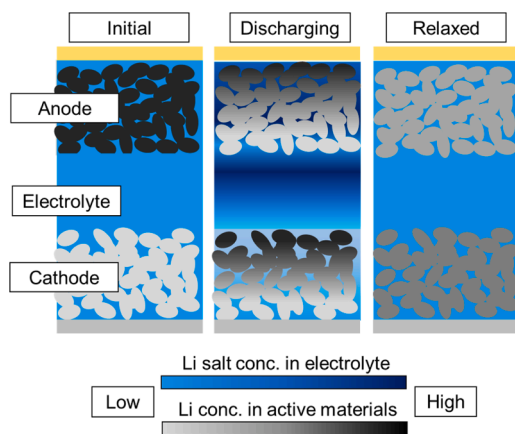


Fig. 5. Schematic illustration of the change of lithium salt concentration in the electrolyte within composite electrodes during high-rate discharging. The initial state shows a uniform salt distribution. During discharging, lithium ions are released from the anode, leading to transient salt accumulation inside the porous electrode and the formation of concentration gradients. After stopping the current, ambipolar diffusion gradually relaxes these gradients and restores a uniform salt distribution within several tens of seconds.

diffusion of individual ions, but by the ambipolar diffusion of electrically neutral lithium salt. In general, the characteristic diffusion length L is given by $L \sim \sqrt{Dt}$, where D is the diffusion coefficient and t is time. The corresponding salt diffusivity in typical LiPF₆-based bulk electrolytes is on the order of $1\text{--}2 \times 10^{-10} \text{ m}^2/\text{s}$, as reported by *operando* NMR [31,32]. Using this relationship, the characteristic diffusion time for a representative length scale of $50 \text{ } \mu\text{m}$ is estimated as $t \sim L^2/D$, yielding

approximately 10–20 s. Such a time scale is in good agreement with the observed relaxation time after discharge and charge shown in Fig. 3c and F, supporting that the observed concentration relaxation is governed by salt diffusion in the electrolyte. Consequently, under high-rate discharge conditions such as 10C or 5C, the time scale of current-induced salt redistribution becomes shorter than the diffusion time scale, leading to persistent salt accumulation inside the electrode. In contrast, near the separator, where tortuosity is lower, concentration relaxation proceeds more smoothly, suppressing large deviations from the initial salt concentration. The slower relaxation within the composite electrode can be attributed to transport limitations arising from the porous electrode structure: electrolyte transport occurs through a tortuous pore network, which reduces the effective diffusion coefficient and restricts transport pathways, resulting in longer diffusion times even over similar length scales. These phenomena are transient; once the electrochemical reaction is stopped, diffusion driven by the concentration gradient restores spatial uniformity within several tens of seconds. Consequently, it is difficult to detect such concentration inhomogeneities after disassembling the battery.

Such salt concentration changes directly affect the ionic conductivity of lithium-ion battery electrolytes. Indeed, the ionic conductivity of typical LiPF₆-based electrolytes exhibits a maximum near an initial salt concentration of approximately 1.0 M and decreases as the concentration deviates from this optimum value [8,36]. Consequently, during fast charge–discharge cycles, the ionic conductivity of the electrolyte is reduced at both the anode and cathode due to local salt accumulation or depletion. This effect is supported by previous half-cell studies using high-concentration electrolytes, in which an increase in salt concentration in the working electrolyte was proposed to improve the effective ionic conductivity at the cathode during discharge [25]. As a result, the discharge capacity was reported to exceed that obtained with a 1.0 M electrolyte. In this context, ion transport and interfacial reactions in typical liquid electrolyte systems are intrinsically coupled, making it challenging to strictly separate the contributions of reaction heterogeneity and diffusion limitations; however, electrolyte stability and transport properties are considered to influence the observed concentration behavior [37]. Our observation results demonstrate that a certain degree of electrolyte concentration variation is an intrinsic and unavoidable phenomenon during high-rate operation of lithium-ion batteries. Since higher current densities induce larger salt concentration gradients, the resulting decrease in ionic conductivity is expected to contribute to increased polarization under high-rate conditions.

4. Conclusion

We developed an *operando* X-ray transmission microscopy method for two-dimensional mapping of electrolyte salt concentration in a lithium-ion battery cell. By using the absorption contrast of PF₆[−], we directly visualized spatiotemporal salt redistribution across the anode and separator under fast charge–discharge conditions. The measurements revealed that significant, rate-dependent salt concentration inhomogeneities form within the anode during fast charge–discharge, while the separator region relaxes more rapidly. These observations indicate that salt transport in the composite electrode is strongly limited by the porous microstructure, leading to transient salt accumulation and depletion that cannot fully relax on the time scale of high-rate cycling. As a result, electrolyte concentration inhomogeneities dynamically form and relax during operation, making them difficult to capture by post-mortem analysis. Because electrolyte ionic conductivity is maximized near 1.0 M and decreases when the concentration deviates from this optimum, such transient concentration inhomogeneity is expected to temporarily reduce local ionic conductivity and increase voltage polarization.

CRediT authorship contribution statement

Hisao Yamashige: Writing – original draft, Investigation, Data curation, Conceptualization. **Noritoshi Furuta:** Investigation, Formal analysis. **Masato Mukoyama:** Investigation, Formal analysis. **Takamasa Nonaka:** Methodology, Investigation. **Yuki Orikasa:** Writing – review & editing, Writing – original draft, Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.electacta.2026.148782](https://doi.org/10.1016/j.electacta.2026.148782).

Data availability

Data will be made available on request.

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